

CHAPTER 13

The Global Cost and Availability of Capital

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Chapter 13

Learning Objectives

01

Explore how corporate strategy and financial globalization may align

03

Evaluate the effect of market liquidity and segmentation on the cost of capital

02

Examine how the CAPM cost of equity changes for the multinational enterprise (MNE)

04

Compare the WACC of an MNE with its domestic counterpart



1

Why Global Capital Markets Matter

Market access, liquidity, and segmentation

The Case for Global Capital Access

Global integration of capital markets gives firms access to cheaper and more abundant funding beyond their home markets.

Problem: Illiquid / Segmented Home Market

- High cost of capital
- Limited availability of long-term funds
- Reduced competitive position
- Constrained investment and growth



Solution: Access to Global Capital Markets

- Lower weighted average cost of capital
- Greater availability of long-term funds
- Stronger competitive position
- Broader investment opportunity set

What Is a Segmented Capital Market?

A national capital market is segmented if the required rate of return on securities differs from comparable-risk securities traded in other markets.

Main Causes of Segmentation

Government constraints (capital controls, taxes)

Institutional practices (accounting differences, disclosure)

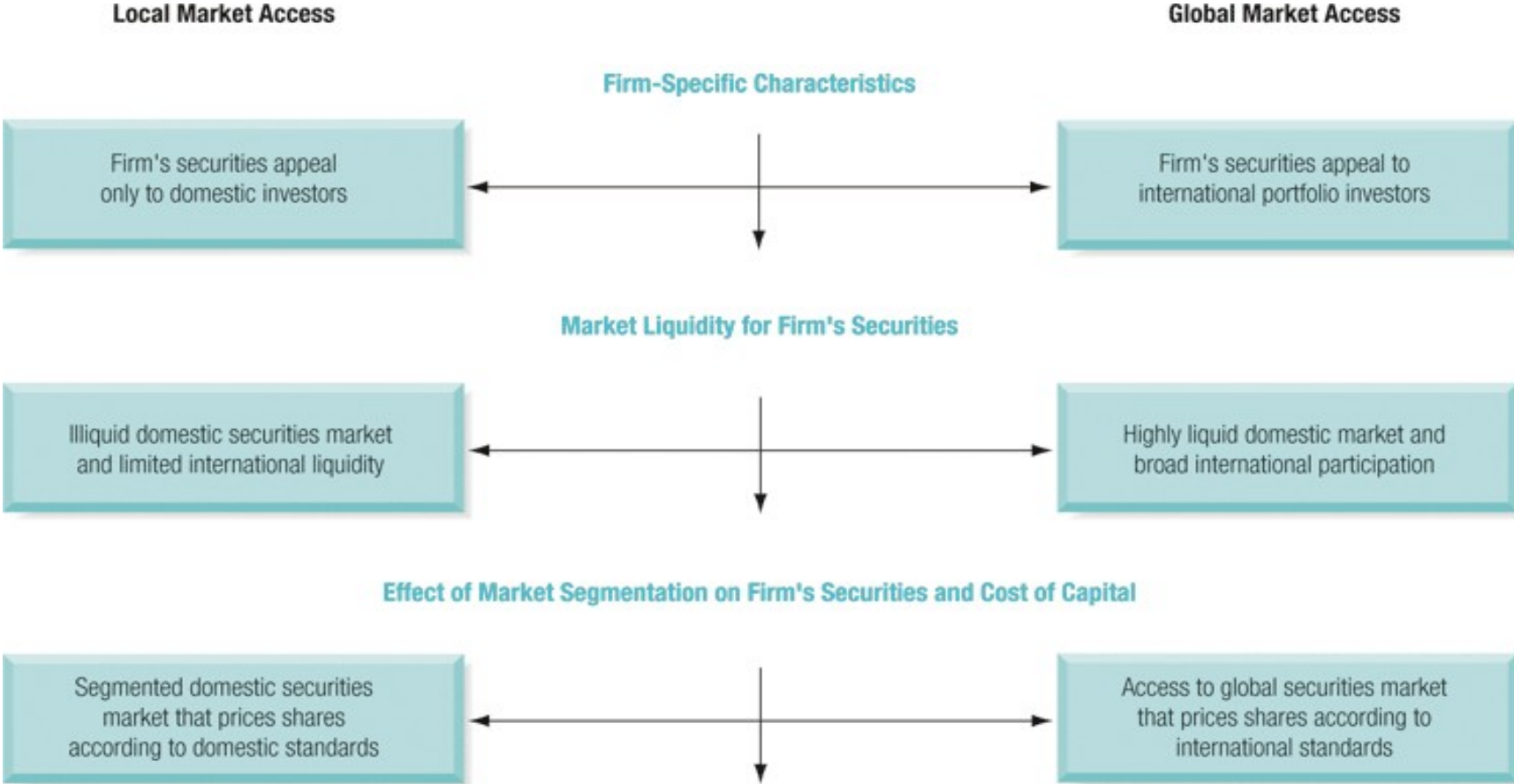
Investor perceptions (home bias, unfamiliarity)

Market Imperfections That Cause Segmentation

- Asymmetric information & lack of transparency
- High transaction costs
- Foreign exchange risks
- Political risks
- Corporate governance issues
- Regulatory barriers

 **Discussion:** *Can a market be domestically efficient but internationally segmented?*

Exhibit 13.1 – Dimensions of the Cost & Availability of Capital Strategy





2

Building Blocks: WACC, Cost of Equity & Beta

The fundamental framework before going global

Cost of Equity — The CAPM Approach

$$\underline{k_e} = \underline{k_{rf}} + \beta \times (\underline{k_m} - \underline{k_{rf}})$$

k_{rf}

Risk-Free Rate

Return on a riskless asset (e.g. US
Treasury bill)

β

Beta

Firm's systematic risk relative to the
market

$(k_m - k_{rf})$

Equity Risk Premium

Extra return investors demand for
taking on market risk

Understanding Beta (β)

$\beta < 1.0$

Less volatile than market (e.g., utilities)

$\beta = 1.0$

Moves exactly with the market (fully diversified domestic portfolio)

$\beta > 1.0$

More volatile than market (e.g., tech startups)

Cost of Debt — $kd(1 - t)$

$$\text{After-tax cost of debt} = \underline{kd} \times (1 - \text{tax rate})$$

How to Calculate $\underline{kd}$

1. Forecast interest rates over the planning horizon
2. Identify the proportions of each class of debt
3. Weight and average the pre-tax interest rates $\rightarrow \underline{kd}$
4. Multiply by $(1 - t)$ to reflect tax deductibility of interest

Worked Example

Debt Structure:

50% Senior bonds @ 5%
30% Subordinated bonds @ 7%
20% Bank loans @ 6%

Weighted $\underline{kd}$:

$$(0.5 \times 5\%) + (0.3 \times 7\%) + (0.2 \times 6\%) = 5.8\%$$

After-tax cost ($t = 25\%$):

$$\underline{kd}(1-t) = 5.8\% \times 0.75 = 4.35\%$$

? Why does the tax shield matter? What happens to $\underline{kd}(1-t)$ if the tax rate rises from 25% to 35%?

Weighted Average Cost of Capital (WACC)

$$\text{WACC} = [k_e \times E / (D + E)] + [k_d(1 - t) \times D / (D + E)]$$

Key Terms

k_e Cost of equity capital

k_d Before-tax cost of debt

t Corporate income tax rate

$E / (D + E)$ Equity as proportion of total capital

$D / (D + E)$ Debt as proportion of total capital

Why does WACC matter?

It is the minimum return a firm must earn on new investments to satisfy all capital providers.

If a project's IRR > WACC → creates value.
If a project's IRR < WACC → destroys value.

? **Quick Check:** If a firm has $k_e = 12\%$, $k_d = 6\%$, $t = 30\%$, and the D/E ratio is 1:1, what is the WACC?



3

International Portfolio Theory & Diversification

How going global changes risk

Portfolio Risk: Domestic vs. International Diversification

Adding assets reduces unsystematic risk, but systematic (market) risk remains. International diversification goes further — national markets are imperfectly correlated.

Domestic Diversification

- Adding more domestic stocks reduces firm-specific (unsystematic) risk
- Systematic risk ($\beta=1.0$) remains — you cannot diversify away the local market's ups and downs
- Returns become correlated with home market

International Diversification

- Adds assets from markets with different economic cycles
- Correlation between national markets is imperfect → more total risk reduction
- BUT: adds currency risk — investor buys two assets: the foreign stock AND the foreign currency

? If you hold a US stock and a Vietnamese stock, and the USD/VND rate swings 10%, how does that affect your portfolio in USD terms?

1. Covariance: Do the variables move together?

The formula is:

$$Cov(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

- **Positive covariance:** X and Y tend to move in the same direction.
- **Negative covariance:** X and Y tend to move in opposite directions.
- **Zero covariance:** No linear relationship.

However, a covariance of **100** or **1000** does not tell you whether the relationship is strong because the number depends on the units.

2. Correlation: How strongly do they move together?

The formula is:

$$\rho_{XY} = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

Correlation takes covariance and **standardizes** it.

As a result:

- **+1:** Perfect positive relationship
- **0:** No linear relationship
- **-1:** Perfect negative relationship

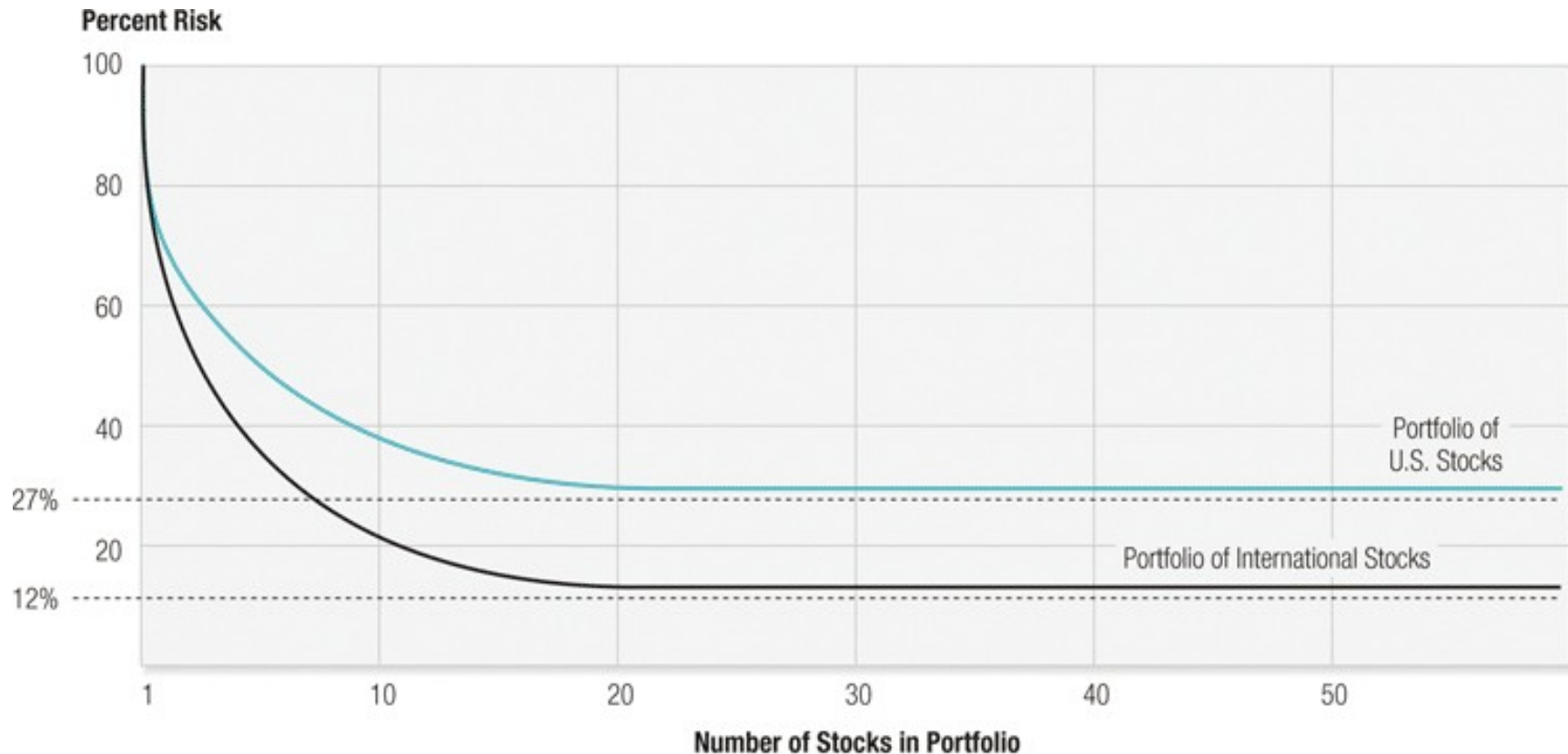
Rule 3: Variance of the rate of return on the portfolio

$$\sigma_p^2 = (w_B \sigma_B)^2 + (w_S \sigma_S)^2 + 2w_B w_S Cov_{BS}$$

- cov_{BS} is the covariance between stock fund and bond fund and $Cov_{BS} = \rho_{BS} \sigma_S \sigma_B$
- So, we can rewrite:

$$\sigma_p^2 = (w_B \sigma_B)^2 + (w_S \sigma_S)^2 + 2(w_B \sigma_B)(w_S \sigma_S) \rho_{BS}$$

Exhibit 13.2 – Gains from Domestic vs. International Diversification



The internationally diversified portfolio achieves lower risk for the same level of return than a purely domestic portfolio.

When the portfolio is diversified internationally, the portfolio's beta—the level of systematic risk which cannot be diversified away—is lowered.



4

International CAPM (ICAPM)

Extending CAPM to global financial markets

From CAPM to ICAPM — What Changes?

ICAPM: assumes financial markets are global — the benchmark portfolio is a world market index, not a domestic one.

Component	Domestic CAPM	ICAPM Adjustment
Risk-free rate (k_{rf})	Domestic government bond yield	Unlikely to change significantly
Beta (β)	Systematic risk vs. domestic market	Recalculated vs. world market index — likely changes, often decreases
Equity risk premium ($k_M - k_{rf}$)	Domestic market excess return	Global market excess return
Cost of equity (k_E)	Domestic CAPM k_E	ICAPM k_E — often lower for MNEs with global exposure

? A Swiss firm moves from domestic to ICAPM. Its domestic β was 1.2. Would you expect the global β to be higher, lower, or the same? Why?

Exhibit 13.3 – Cost of Equity for Nestlé of Switzerland

Nestlé's estimate of its cost of equity will depend upon whether a Swiss investor is thought to hold a domestic portfolio of equity securities or a global portfolio.

Domestic Portfolio for Swiss Investor

$k_{RF} = 3.3\%$ (Swiss bond index yield)

$k_M = 10.2\%$ (Swiss market portfolio in SF)

$\beta_{Nestlé} = 0.885$ (Nestlé versus Swiss market portfolio)

Global Portfolio for Swiss Investor

$k_{RF} = 3.3\%$ (Swiss bond index yield)

$k_M = 13.7\%$ (*Financial Times* Global index in SF)

$\beta_{Nestlé} = 0.585$ (Nestlé versus FTA-Swiss index)

$$k_{Nestlé} = k_{RF} + \beta_{Nestlé}(k_M - k_{RF})$$

Required return on Nestlé:

$$k_e^{Nestlé} = 9.4065\%$$

Required return on Nestlé:

$$k_e^{Nestlé} = 9.3840\%$$

Source: All values are taken from René Stulz, "The Cost of Capital in Internationally Integrated Markets: The Case of Nestlé," *European Financial Management*, Vol. 1, No. 1, March 1995, pp. 11–22.

Notice: Nestlé's ICAPM cost of equity is lower than the domestic CAPM estimate — reflecting its access to global capital and reduced systematic risk when measured globally.



5

The Equity Risk Premium Debate

Why practitioners can't agree — and why it matters

Equity Risk Premium (ERP) — What It Is and Why It's Controversial

ERP = $k_m - k_{rf}$ · The average annual return the market is expected to earn above riskless debt



Widely Agreed

CAPM is the preferred method for estimating cost of equity, and the estimate should be forward-looking.



Contested

What numerical value to use for ERP?
Arithmetic vs. geometric averages produce different results. Estimates differ across countries and time periods.



Common Practice

Practitioners typically use historical data as a basis for forward-looking projections — but even the historical record is debated.

When discount rates differ, project valuations differ — dramatically. The ERP debate isn't academic; it directly affects capital budgeting decisions.

Why is this important in finance?

Suppose an investment returns:

- Year 1: +50%
- Year 2: -50%

Arithmetic average return:

$$\frac{50\% - 50\%}{2} = 0\%$$

Geometric average return:

$$\sqrt{(1.5)(0.5)} - 1 = \sqrt{0.75} - 1 = -13.4\%$$

Although the arithmetic average says the average return is 0%, your wealth actually falls:

$$100 \rightarrow 150 \rightarrow 75$$

So the **geometric average is lower because volatility reduces compound growth**. This is why finance uses:

- **Arithmetic average** → expected one-period return.
- **Geometric average** → actual long-term compound growth rate.

Exhibit 13.4 – How Different ERP Assumptions Drive Different k_e Estimates ($\beta = 1$, $k_{rf} = 4\%$)

Source	Equity Risk Premium ($k_m - k_{rf}$)	Cost of Equity $k_{rf} + \beta (k_m - k_{rf})$	Differential
Ibbotson	8.800%	12.800%	3.800%
Finance textbooks	8.500%	12.500%	3.500%
Investor surveys	7.100%	11.100%	2.100%
Dimson, et al.	5.000%	9.000%	Baseline

Source: Equity risk premium quotes from "Stockmarket Valuations: Great Expectations," *The Economist*, January 31, 2002.

Key insight: With $\beta = 1$ and $k_{rf} = 4\%$, choosing an ERP of 3.5% vs. 7.5% changes k_e from 7.5% to 11.5% — a 4 percentage point swing that can dramatically alter NPV calculations.



6

International Portfolio Investors & Market Liquidity

Who buys foreign securities, and why

What Motivates International Portfolio Investors?

Deregulation of equity markets over the past three decades has opened markets to foreign investors. Both domestic and international managers aim to maximise return for a given risk — or minimise risk for a given return.



Portfolio Risk Reduction

National markets are imperfectly correlated → adding foreign assets reduces portfolio risk below what domestic-only diversification achieves.



Higher Expected Returns

International managers choose from a larger asset bundle → often achieve higher risk-adjusted returns than domestic-only managers.



Currency Risk

Buying a foreign stock means acquiring two assets: the security itself AND the foreign currency. Exchange rate swings can amplify or reduce returns.

? A portfolio manager holds 60% US stocks and 40% emerging market stocks. If EM currencies depreciate 15% against the USD, does the EM allocation help or hurt overall portfolio risk?

How Market Liquidity Affects the Cost of Capital

Market liquidity = the degree to which a firm can issue new securities without depressing the existing market price.
Higher liquidity → lower marginal cost of capital.

Domestic Market Only

- As the firm raises more capital, it saturates the domestic market.
- Investors demand a higher return to absorb the additional supply.
- Marginal cost of capital (MCC) rises steeply.
- Result: smaller optimal capital budget.

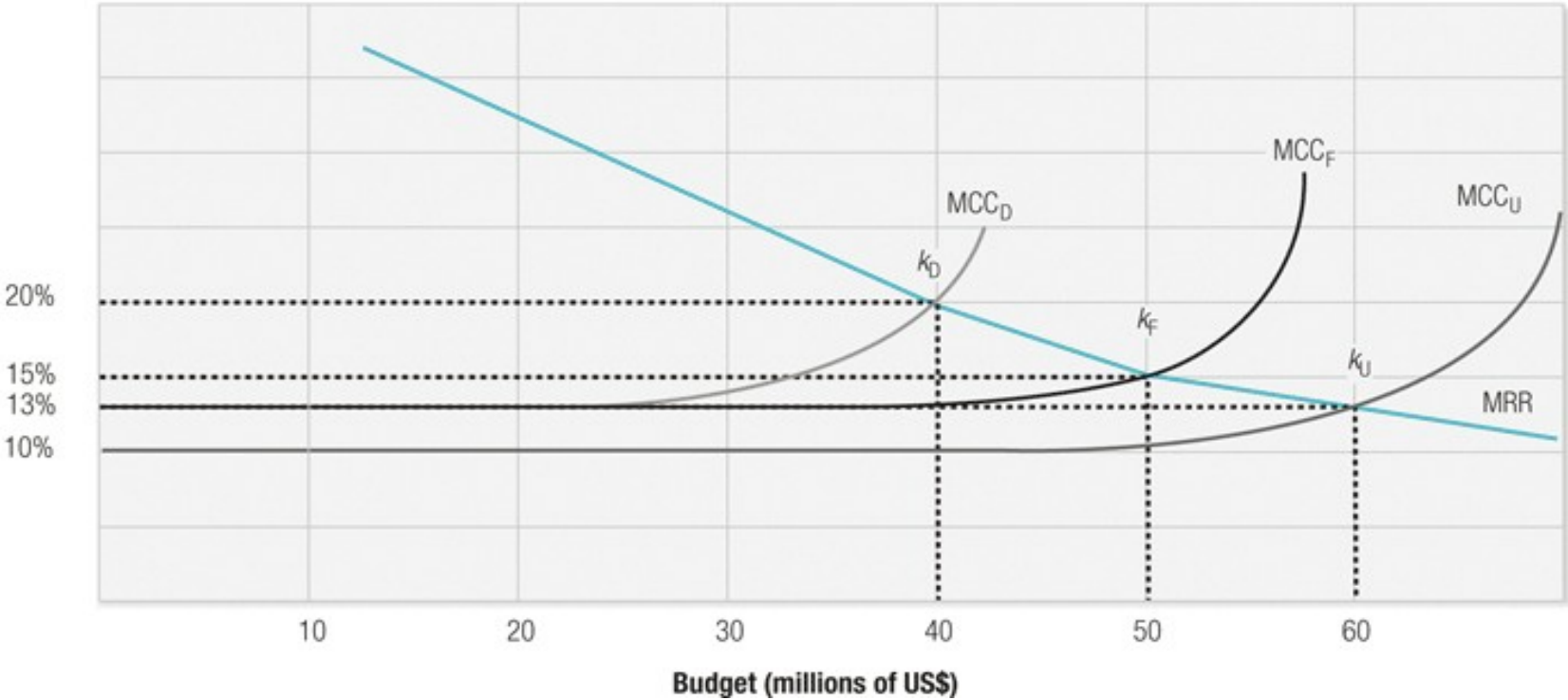


Global Capital Markets Access

- Firm taps multiple markets — domestic saturation is avoided.
- MCC curve shifts rightward — firm can raise more capital at the same (or lower) cost.
- For segmented markets: access to global equity markets further reduces MCC.
- Result: larger optimal capital budget → more investment.

Exhibit 13.5 – Market Liquidity, Segmentation & the Marginal Cost of Capital

Marginal Cost of Capital
and Rate of Return (percentage)



MCC_D = domestic (illiquid) · MCC_F = with access to foreign capital · MCC_U = from segmented market gaining global access · MRR = marginal rate of return on projects



7

MNE vs. Domestic Firm Cost of Capital

Does going global actually lower your WACC?

Theory vs. Reality: Does the MNE Have a Lower WACC?

Theory Says: MNE Should Have Lower WACC

- Access to global capital → lower marginal cost of capital
- International diversification reduces systematic risk
- Larger opportunity set → more value-creating projects



Empirical Reality: MNEs Often Have Higher WACC

- Additional political risk in foreign operations
- Agency costs of managing across borders
- Currency risk adds to required returns
- Complexity and information asymmetry
- Expanded opportunity set → capital budget rises beyond the 'flat' part of the MCC curve

? Does this mean MNEs are worse businesses? Or is the higher WACC compensated by higher project returns?

The Link: Opportunity Set, Capital Budget, and WACC

As the opportunity set expands, the firm raises more capital, eventually moving to a region where the MCC is rising. The optimal budget sits where $MCC = MRR$.

1

Larger opportunity set

MNE has more global projects, so its MRR curve (declining) sits above that of a domestic firm.

2

Bigger capital budget

To exploit those projects, the MNE raises more capital — pushing further along the rising MCC curve.

3

Higher WACC

At the new optimal budget, MCC is higher. So the MNE's WACC is higher — even though the MCC curve itself may not be steeper.

4

Still creates more value

If MRR at the expanded budget exceeds the higher MCC, the MNE still generates more value than a purely domestic firm.

Exhibit 13.6 – MNE vs. Domestic Firm: Cost of Capital Compared

Marginal Cost of Capital
and Rate of Return (percentage)

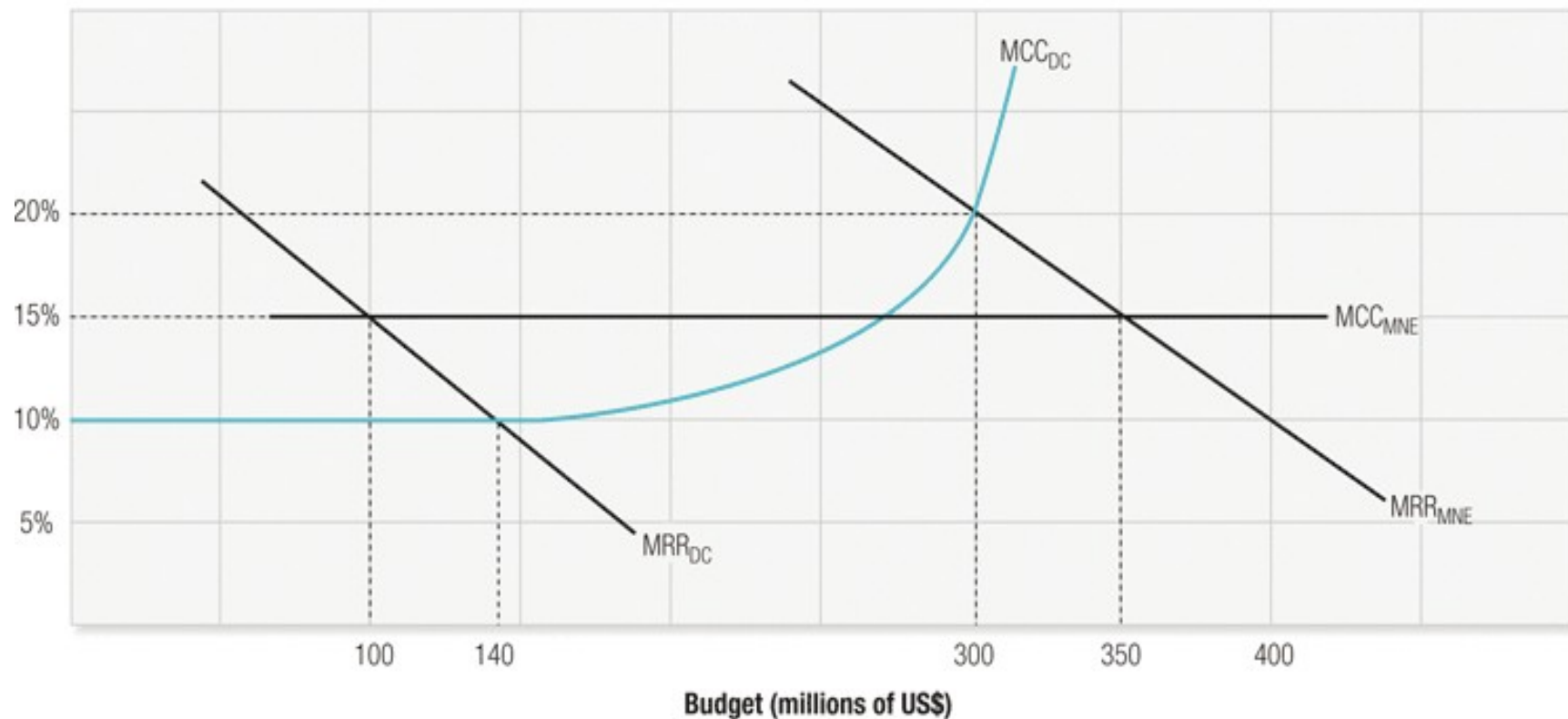


Exhibit 13.7 – Do MNEs Have Higher or Lower WACC? Summary

Is $MNE_{WACC} > \text{or} < Domestic_{WACC}$?

$$k_{WACC} = k_e \left[\frac{\text{Equity Value}}{\text{Value}} \right] + k_d(1-t) \left[\frac{\text{Debt Value}}{\text{Value}} \right]$$

↓

Empirical studies indicate that MNEs have a lower debt/capital ratio than domestic counterparts, indicating that MNEs have a higher cost of capital.

↓

And indications are that MNEs have a lower average cost of debt than domestic counterparts, indicating that MNEs have a lower cost of capital.

↓

The cost of equity required by investors is higher for multinational firms than for domestic firms. Possible explanations are higher levels of *political risk*, *foreign exchange risk*, and higher *agency costs* of doing business in a multinational managerial environment. However, at relatively high levels of the optimal capital budget, the MNE would have a lower cost of capital.

Conclusion: Empirically, MNEs have higher WACC. But if they also limit their budget to where MCC is still flat, the gap shrinks. The real advantage is the volume of value-creating investment, not a lower cost of capital.



8

Review & Practice Questions

Applying the concepts

Mini Case: Should Vinamilk Cross-List on a Global Exchange?

Vinamilk is a large Vietnamese consumer firm currently listed only on HoSE. Management is considering a cross-listing in Singapore or Hong Kong.

Arguments For Cross-Listing

- Access to larger pool of international investors
- Potential reduction in k_e via lower global β
- Stronger corporate governance signals $\rightarrow$ lower required return
- Ability to issue more equity without depressing HoSE price

Arguments Against / Risks

- Listing costs (fees, legal, disclosure)
- Stricter reporting requirements
- Additional forex risk for foreign shareholders
- Risk that cross-listing does not achieve expected price uplift

? Group Task: What data would you need to decide? How would you quantify the cost of capital benefit vs. the listing cost?

Practice: Applying WACC and CAPM in Global Contexts

Q1

A Vietnamese firm ($k_{rf}=6\%$, $\beta=0.9$, $ERP=8\%$) lists its shares on the London Stock Exchange. Its global β falls to 0.65 and the global ERP is 5.5%. How does k_e change?

Hint: Listing saves ~3.6% on cost of equity.

Q2

A MNE has $k_e=10\%$, $k_d=5\%$, $t=25\%$, and uses 40% debt/60% equity. What is the WACC? If it moves to 50/50, what happens to WACC?

Hint: More debt (within limits) lowers WACC via tax shield.

Q3

Firm A (domestic) has a WACC of 8% and optimal budget of \$200M. Firm B (MNE) has WACC of 9% and optimal budget of \$500M. Which creates more value, assuming the same project quality?

Hint: Firm B likely creates more value despite higher WACC because it deploys far more capital at returns above its hurdle rate.

Chapter 13 — Key Takeaways

1

Firms in illiquid/segmented markets face higher cost of capital → escaping via global listing lowers k_e and WACC.

2

WACC = weighted cost of equity (CAPM/ICAPM) + after-tax cost of debt. Every component can change when the firm goes global.

3

International diversification reduces portfolio risk below domestic levels, but introduces currency risk.

4

ICAPM replaces the domestic beta with a global beta — often lower — and a global equity risk premium.

5

The ERP is the most contested input in CAPM: arithmetic vs. geometric averages, historical vs. forward-looking, vary across markets.

6

Empirically, MNEs have higher WACC than domestic peers — but this is partly because they invest more, not because global capital is expensive.